# Grant Proposals Evaluation Report

## Summary of Proposal Scores

| Proposal | Score | Rank |
|---|---|---|
| AI Enabled Portfolio Management.pdf | 8.1 | Excellent |
| AI Enabled Source Selection for Contract Proposal Evaluation.pdf | 8.2 | Excellent |
| AI_ML-Generated Decoy Networks.pdf | 6.7 | Good |
| Novel AI Techniques for Insights in Various Environments.pdf | 8.2 | Excellent |

# Proposal: AI Enabled Portfolio Management.pdf

**Overall Score:** 8.1

**Overall Rank:** Excellent

### *COMPANY PROPOSAL*

**Score:** 7.2

**Reason:** The proposal provides a clear overview of the project scope, timelines, and technical approach.

**Suggestions:** Consider adding more detail on how the AI-powered platform will address specific pain points in the Army's acquisition process and provide evidence of successful testing and validation of the prototype.

### *performance*

**Score:** 8.2

**Reason:** Proposal provides a clear plan for achieving performance milestones in Phase II, with detailed timelines and deliverables. However, some of the technical objectives and deliverables could be more specific and measurable.

**Suggestions:** Consider providing more quantitative targets for each milestone, such as percentage increase in AI-driven acquisition efficiency or number of contracts matched accurately within a certain timeframe.

### *requirements approval*

**Score:** 8.5

**Reason:** The proposal effectively addresses the requirements approval section by providing a clear plan for development of modern Acquisition requirements intake and portfolio management capability leveraging AI capabilities and commercial market research R&D.;

**Suggestions:** None

### *Section 2*

**Score:** 8.5

**Reason:** The proposal adequately addresses the challenges in Acquisition requirements intake and portfolio management, providing a clear and structured plan for development of an AI-Enabled Portfolio Management platform.

**Suggestions:** Consider incorporating more specific metrics and benchmarks to evaluate the success of the proposed solution and its alignment with Army's priorities and standards.

### *Phase II*

**Score:** 8.5

**Reason:** The proposal provides a clear and detailed plan for the initial MVP and subsequent MVPs, outlining key technical objectives, timelines, and deliverables. However, some aspects of the Phase II approach could be improved, such as providing more specific metrics for success and a clearer description of how the AI framework will be adapted for enterprise-wide Army rollout.

**Suggestions:** Provide more specific metrics for success, such as quantifiable targets for accuracy, latency, and throughput. Clarify the process for adapting the AI framework for enterprise-wide Army rollout.

### *Phase III Dual Use Applications*

**Score:** 8.0

**Reason:** The proposal provides a clear understanding of the need for an AI-enabled portfolio management solution tailored to Federal Acquisition processes and regulations.

**Suggestions:** The proposal could benefit from more detailed explanations of how the proposed solution addresses specific dual-use opportunities in finance, healthcare, and other highly regulated sectors.

### *Framework for Developing an AI-Enabled Portfolio Management Tool*

**Score:** 8.0

**Reason:** The proposal demonstrates a clear understanding of the importance of developing a compliant portfolio management tool applicable to other highly regulated industries.

**Suggestions:** To further strengthen this section, consider highlighting specific examples or case studies where similar frameworks have been successfully applied in Finance and Healthcare.

## Proposal: AI Enabled Source Selection for Contract Proposal Evaluation.pdf

**Overall Score:** 8.2

**Overall Rank:** Excellent

### *COMPANY PROPOSAL*

**Score:** 8.5

**Reason:** The proposal provides a clear and detailed technical approach, including Phase I equivalency evidence, key personnel, company overview, facilities & environment, work plan & milestones, and phase III & dual-use applications. However, some of the language used is repetitive or overly generic, which may detract from the overall impact of the proposal.

**Suggestions:** Consider condensing repetitive information, using more precise and technical language to describe system components and processes, and providing additional details on the AI/rule-engine architecture.

### *Phase I*

**Score:** 8.0

**Reason:** The proposal provides a clear and detailed explanation of the Phase I equivalent evidence required to substantiate scientific and technical merit and feasibility.

**Suggestions:** Ensure that all necessary documentation is attached or linked to the proposal.

### *Section 2*

**Score:** 8.5

**Reason:** The proposal addresses the section well by providing a clear and detailed explanation of how the solution will address the evaluator inconsistencies and deviation from solicitation instructions and evaluation criteria.

**Suggestions:** Consider providing more specific examples or case studies to demonstrate the effectiveness of the proposed solution in addressing these challenges.

### *Evaluation of System*

**Score:** 7.8

**Reason:** The proposal provides a clear and well-structured plan for the development of SelectAI-SS, addressing key aspects such as data curation, AI scoring, and blockchain-based audit trails. The Phase II timeline is ambitious but achievable, with clear milestones and deliverables.

**Suggestions:** Consider providing more detailed risk assessments and mitigation strategies for complex tasks, and ensure that the team has sufficient expertise in areas such as human-centered design and DevSecOps.

### *Section 5*

**Score:** 8.2

**Reason:** The proposal demonstrates a clear understanding of the urgent need for rapid delivery and the importance of demonstrating expert-level knowledge of Army acquisition processes, standards, and practices. However, some details regarding the deliverables and timelines could be clarified or expanded upon.

**Suggestions:** Provide more specific examples of how SelectAI-SS will adapt to meet the evolving needs of the Army and other federal agencies; consider providing a detailed Gantt chart or project schedule to ensure clear understanding of the timeline and milestones.

### *Phase III Dual-Use Applications*

**Score:** 8.5

**Reason:** The proposal demonstrates a clear understanding of the unique challenges in automating the source selection process for federal agencies and provides a solid foundation for developing an AI-enabled portfolio management tool compliant with Federal Acquisition processes and regulations.

**Suggestions:** Consider providing more specific details on how SelectAI-SS can be tailored for other federal agencies and highly regulated sectors, as well as potential partnerships or collaborations that can further enhance the solution.

### *Framework for developing an AI-enabled portfolio management tool*

**Score:** 8.2

**Reason:** The proposal provides a clear and well-defined framework for the development of an AI-enabled source selection tool, with a focus on compliance with federal regulations.

**Suggestions:** Consider providing more detail on how the proposed framework will be applied in other highly regulated industries such as Finance and Healthcare.

# Proposal: AI_ML-Generated Decoy Networks.pdf

**Overall Score:** 6.7

**Overall Rank:** Good

## *Company Proposal*

**Score:** 8.2

**Reason:** The proposal provides a clear overview of the project objectives, technical approach, and key personnel. However, some aspects could be improved for better alignment with the call section.

**Suggestions:** Clarify how the company's experience in delivering ML-driven defense solutions aligns with the proposed technology and its relevance to the AF254-0801 topic. Enhance the feasibility study report by providing more details on data requirements and AI/ML approaches evaluated.

## *Section 2*

**Score:** 6.5

**Reason:** The proposal addresses the need for a more robust defense against state-sponsored hackers by proposing a novel approach using AI/ML techniques to generate realistic decoy networks. However, it lacks specificity on how the proposed solution will comply with export control regulations and the use of foreign nationals.

**Suggestions:** Please provide detailed information on how DecoyGen-AI will comply with US Export Control Laws and specifically address the risks associated with using foreign nationals in the development process.

## *Phase 1 Section*

**Score:** 7.5

**Reason:** Proposal provides a comprehensive feasibility study with AI/ML methodologies, data pipeline design, and modular software architecture. However, it may benefit from more detailed analysis of current methods for decoy realism and adaptability.

**Suggestions:** Consider adding more specific examples or case studies to demonstrate the effectiveness of selected AI/ML approaches in generating realistic decoy networks.

## *Section 3*

**Score:** 0.0

**Reason:** {"section": "Phase I Proposal", "score": 8.2, "reason": "The proposal demonstrates a good understanding of the topic and outlines a clear plan for addressing the call section on anomaly/intrusion detection software and decoy networks. However, some aspects of the proposal could be improved upon to make it more competitive.", "suggestions": "Consider adding more specific details on how the proposed solution addresses the limitations of current decoy networks and provides novel approaches to create realistic "digital twin" decoy networks. Additionally, provide more information on how the AI/ML techniques will be integrated with blockchain-backed state management and expert "white-hat" feedback loops."}

**Suggestions:**

## *PHASE I*

**Score:** 8.2

**Reason:** The proposal provides a clear outline of the approach, methodology, and technical objectives for Phase I. The team has also demonstrated expertise in AI/ML and cybersecurity through their backgrounds. However, some minor concerns exist regarding the scope and deliverables.

**Suggestions:** Consider expanding on the technical objectives and providing more detail on how each objective will be addressed. Additionally, ensure that the phase I deliverables align with the overall project goals and that the feasibility study report is comprehensive.

## *PHASE II*

**Score:** 8.0

**Reason:** The proposal adequately addresses Phase II requirements by providing a realistic development sandbox environment for demonstrating the prototype at TRL 6 maturity.

**Suggestions:** To further enhance the proposal, consider providing more details on how Risk Management Framework integration will be achieved in Phase II.

## *Section 7*

**Score:** 8.2

**Reason:** The proposal addresses the requirement for addressing DoD, USG, and commercial markets by mentioning market expansion beyond the current DoD client base and potential partnership with other government agencies or private sector entities to leverage DecoyGen-AI.

**Suggestions:** Consider providing more specific details on how DecoyGen-AI will be adapted to meet the unique needs of each market segment (e.g., DoD, USG, commercial). Additionally, outline a plan for ensuring the commercialization and scalability of the technology.

# Proposal: Novel AI Techniques for Insights in Various Environments.pdf

**Overall Score:** 8.2

**Overall Rank:** Excellent

### *Section 1*

**Score:** 8.2

**Reason:** The proposal clearly states the problem and objectives of the project, but could benefit from more detailed explanations of the proposed AI techniques and their relevance to the specific application.

**Suggestions:** Provide a clearer explanation of how self-supervised vision transformers, foundational multi-modal models, generative anomaly synthesis, and retrieval-augmented generation will be used to address the problem, and provide more context about the specific datasets and use cases being addressed.

### *Phase II Technical Objectives*

**Score:** 8.5

**Reason:** The proposal outlines specific objectives for Phase II, including refining AI models, deploying a robust and secure prototype, integrating with Project Linchpin, and performing operational validation.

**Suggestions:** Consider adding more detail on how the proposed technical objectives will be achieved, such as high-level descriptions of the methodologies and timelines for each objective.

### *Phase II Technical Objectives*

**Score:** 9.0

**Reason:** The proposal clearly outlines the technical objectives for Phase II, including refining and scaling AI models, robustly deploying a secure prototype, integrating with Project Linchpin, and conducting operational validation.

**Suggestions:**

### *Phase II*

**Score:** 8.5

**Reason:** The proposal provides a detailed plan for the Phase II project, including specific technical objectives, timelines, and deliverables.

**Suggestions:** The proposal could benefit from more emphasis on the overall project timeline, with clearer milestones and dependencies between each phase.

### *marketplace*

**Score:** 8.0

**Reason:** The proposal addresses the technical requirements for IL5-compliant deployment and showcases a clear understanding of AI model refinement and generative anomaly expansion. However, the extent to which these elements contribute to performance, accuracy, and security standards may require further explanation or validation.

**Suggestions:** Consider providing more detailed analysis on how the proposed models meet performance, accuracy, and security standards, such as providing metrics on recall, precision, F1 score, and ROC-AUC, as well as explanations of the generative anomaly expansion approach.

### *Phase III Dual-Use Applications*

**Score:** 8.2

**Reason:** The proposal provides a clear description of the dual-use applications for the AI technology, but could benefit from more concrete examples and specific metrics to demonstrate the impact of these applications.

**Suggestions:** Consider providing more detailed case studies or success stories for each application area, as well as specific metrics or results that demonstrate the effectiveness and efficiency of NATIVE-Insight in addressing these challenges.

### *Section 7*

**Score:** 9.0

**Reason:** The proposal adequately addresses the need for AI in various industries, providing a comprehensive overview of potential applications.

**Suggestions:**

### *Section 8*

**Score:** 6.0

**Reason:** The proposal effectively showcases the versatility of AI in various industries beyond defense, demonstrating its potential for increased productivity, competitiveness, and economic growth.

**Suggestions:** Consider adding more specific examples or case studies from these industries to further illustrate the applicability of NATIVE-Insight.